

AgentPay × Substreams

Architecture Update for The Graph

From: Shawn Lippert, AgentPay | May 19, 2026

Re: Substreams integration status + questions

What We Shipped

The AgentPay Substreams package is **compiled, packed, and streaming live on Base mainnet** as of today.

- **Package:** agentpay-substreams-v0.1.0.spkg
- **Network:** base-mainnet via mainnet.eth.streamingfast.io:443
- **Status:** Live — streaming confirmed on mainnet today, cursor-based, auto-reconnects on drop

Current Pipeline (What's Running Now)

Base L2 Blocks (via Firehose, --final-blocks-only)

```
|
v
[map_usdc_transfers]
- USDC: 0x833589fCD6eDb6E08f4c7C32D4f71b54bdA02913
- TransferWithAuthorization (EIP-3009)
- Topic: 0x98de503528ee59b575ef0c0a2576a82112635b28b9da4cae03c8b0b0ec06b8e3
- Extracts: from, to, value, nonce, block_num, timestamp, tx_hash
|
v
[map_confirmed_settlements] <- final blocks only; all modules below inherit finality
- Filters to transfers TO AgentPay facilitator wallet
- Emits SettlementEvent (payment_id from EIP-3009 nonce)
- Facilitator acts after finality confirmed -- no premature release
|
+----> [store_agent_stats]   StoreAddInt64 with deltas
                        Single store, keyed by agent wallet
                        Tracks: total, success, refunded, unique_counterparties
```

Reputation scoring + analytics sink: next phase (questions below).

Why Substreams vs. Our Old Polling Loop

Old approach: eth_getLogs every 2 seconds.

- 2-4 second lag per payment, manual reorg handling
- ~\$0.002/req x high frequency = real ongoing cost
- Single RPC provider = single point of failure

With Firehose-backed Substreams, all modules run with --final-blocks-only:

- Reorg-safe by design — finality propagates through entire module graph
 - One persistent gRPC stream, no per-request cost
 - Server load stays flat — integrators query the subgraph, not our facilitator
-

Three Things We Need from You

1. x402GrantRegistry — address + startBlock on Base

A registry contract tracking grant-funded x402 payments on Base. We want to add it as a module so grant-funded settlements are distinguishable from standard ones. Without the address we can't build it.

2. Reputation → subgraph or Clickhouse — what's the right pattern?

We want external agents to query `agentReputation(wallet: "0x...")`. We were leaning toward a subgraph for the GraphQL interface, with `store_agent_stats` deltas feeding into a subgraph handler. But we haven't deployed a subgraph before — is that the right shape for this use case, or would you structure it differently?

3. Subgraph deployment path

Starting fresh, no existing subgraph. Hosted service first or straight to the decentralized network?

The .spkg is packed and streaming. Happy to drop the repo link, walk through module code, or coordinate a PR review.

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